

Synthesis, Characterization and Battery Performance of A Lithium Poly (4-vinylphenol) Phenolate Borate Composite Membrane



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ABSTRACT

We report synthesis of lithium poly (4-vinylphenol) phenolate borate (LiPVPPB) single-ion conductor comprised of boron atoms with sp³ electronic configuration covalently bonded to a polystyrene backbone with high thermal and electrochemical stability. The highly delocalized anionic charges surrounding the boron atoms in the polymer give rise to weak association with lithium ions in the polymer matrix, resulting in an ion transference number close to unity and remarkably high ionic conductivity. A composite membrane blended with LiPVPPB and poly(vinylidene-fluoride-co-hexafluoropropene) (PVDF-HFP) was fabricated. The battery of the electrolyte displays excellent cyclability with nearly 100% coulombic efficiency over a wide temperature range. The superior membrane performance suggests that single ion polymer electrolyte materials are highly promising for safe and high power applications of lithium ion batteries.

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1. Introduction

One of the essential components of rechargeable lithium ion batteries is the electrolyte that mediates ion transport between electrodes in the devices.[1] With the rapidly rising demand for lithium ion batteries for applications in portable electronics and more recently in electric vehicles low cost, long cycle life and reliable electrolytes for battery cells have become one of the focal points of intense research activities in recent years.[2,3] The development is chiefly propelled by the fact that conventional liquid electrolytes based on small inorganic salts suffer from the potential safety related issues arising from flammable organic solvents used in a large quantity in battery cells.[4,5] Furthermore, with the recent exciting development of high discharge capacity multicomponent oxide materials for cathode, a lack of adequate electrolytes that match up with the high voltage requirement has become the bottleneck that prevents battery cells from full utilization of the potentially high power density enabled by these cathode materials.[6] Although conventional gel polymer electrolytes, comprised of inorganic lithium salts blended with polymer network, utilize much less organic solvents and thus are much safer, the relatively low ionic conductivity arising from the low ion transference

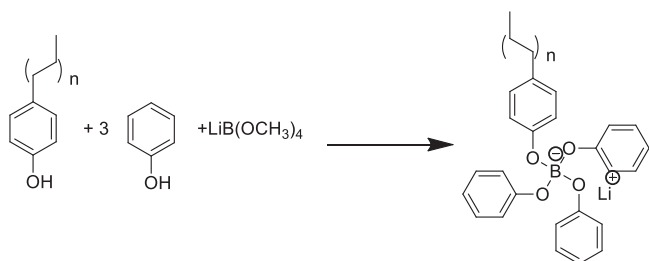
number and the low diffusivity still limits the broad applicability of the gel polymer electrolyte based batteries.[7] It remains to be demonstrated for the current dual-ion based electrolyte materials, including liquid electrolytes and gel polymer electrolytes, to meet the required voltage tolerance and safety standard for high power battery applications.

Single ion polymer electrolytes (SIPEs) represent a new class of electrolytes that differ from dual-ion electrolytes in several aspects. The anions in the SIPE materials all reside on polymer frameworks with the anions surrounded by electron withdrawing groups and are thus immobile during charge and discharge processes; the lithium ions are weakly associated with the anionic polymeric framework with the minimum concentration gradient leading to a substantially higher ion transference number than what is observed in dual-ion based electrolytes.[8–16] In addition to the advantage of the improved ion conduction, the SIPE membranes have been shown to display excellent thermal and mechanical stability and a wide electrochemical window.[17–23] Therefore, these materials may be well suited for applications in high power Li-ion batteries.

Since the pioneer work by Gores et al. on small molecular lithium chelatorborates,[24–29] a number of electrochemically active new boron complexes have been synthesized,[30–36] among which lithium bis(oxalato)borate (LiBOB) and lithium oxalyldifluoroborate (LiODFB) are the most prominent. These compounds exhibit superior thermal stability with a higher lithium ion transference number than conventional liquid organic electrolytes. To minimize the concentration gradient inherently associated with

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Scheme 1. synthesis procedure of LiPVPPB.

these dual-ion based electrolytes, single-ion polymeric electrolytes were proposed. In these materials, anions are chemically anchored in polymer networks and normally delocalized through bonding with electron withdrawing groups. Consequently, the electrostatic interaction between lithium ions and their counter ions becomes weakened, leading to significantly higher Li^+ mobility and a substantially higher ion transference number close to unity. A number of sp^3 boron based single ion electrolytes were successfully synthesized and well characterized.[37–39] However, the ionic conductivity of the early development of single-ion polymer electrolytes was relatively low, only at the level of $10^{-5} \text{ S cm}^{-1}$ at ambient temperature, and the lithium ion transference number was far less than unity. The sp^3 boron based single-ion polymer electrolytes with high ionic conductivity and high lithium ion transference number have only been developed recently.[20,21,40,41]

In a previous communication[41], we reported three sp^3 boron based single-ion electrolytes through polymerization of silylated carboxylic acids with lithium tetramethanolateborate in dehydrated N, N-dimethylformamide. The silylation procedure was conducted under a stringent preparation condition requiring anhydrous of hexamethyldisilazane and 1, 2-dichloroethane prior to use. We recently developed a new synthetic protocol to produce sp^3 boron based polymer electrolytes through direct polymerization using phenols and lithium tetramethanolateborate,[42] which makes the synthetic procedure much simpler. In the present study, we utilized the protocol to synthesize a new sp^3 based single-ion electrolyte through polymerization with poly (4-vinylphenol), lithium tetramethanolateborate and phenol. Subsequently, an electrolyte membrane was fabricated via the doctor-blading method upon blending the synthesized polymer with PVDF-HFP. The physical and electrochemical properties of the membrane were characterized based on a range of spectroscopic experiments. The membrane was then used in a coin cell to test its battery performance. It was demonstrated that the battery performs remarkably well in the temperature range of the test between 25°C and 80°C . The phenol rings connected to the sp^3 boron atoms facilitate charge delocalization via π conjugation around the anionic sites and thus boost the mobility of Li^+ -ions. The large molecular weight of the starting material, poly(4-vinylphenol) ($M_w = 25,000 \text{ g mol}^{-1}$), enables the anions to be immobilized in the charge and discharge processes and thus concentration polarization intrinsically associated with dual-ion based electrolytes is minimized.

2. Experimental

The synthetic procedure is shown in Scheme 1.

2.1. Materials

Poly (4-vinylphenol), $M_w = 25,000 \text{ g mol}^{-1}$ (Sigma-Aldrich), phenol (Sigma-Aldrich), lithium metal (chemmetall), trimethyl

borate (Alfa-Aesar), anhydrous N, N-dimethylformamide (DMF), (Sigma-Aldrich), acetonitrile (Tedia) were used as purchased. Methanol (Fisher) was dried before use. All reagents used were of reagent grade.

2.2. Synthesis and characterization of lithium poly (4-vinylphenol) phenolate borate (LiPVPPB)

2.2.1. Synthesis of lithium tetramethanolateborate ($\text{LiB}(\text{OCH}_3)_4$)

Lithium tetramethanolateborate was synthesized and purified according to the procedure given by Wuhr.[43] NMR spectra in DMSO-d_6 : ^1H , 3.16 ppm (s); ^{13}C , 48.50 ppm; ^{11}B (relative to $\text{BF}_3\cdot\text{Et}_2\text{O}$), 3.62 ppm.[28]

2.2.2. Synthesis of lithium poly (4-vinylphenol) phenolate borate (LiPVPPB)

10 mL of anhydrous N, N-dimethylformamide was added into the mixture of poly (4-vinylphenol) (1.2015 g), phenol (2.8233 g) and lithium tetramethanolateborate (1.4189 g). The mixture was stirred at 50°C for overnight and then at 100°C for 3 days. Upon cooling to room temperature, the solution was poured into 100 mL of diethyl ether. The precipitate was collected upon vacuum filtration and further washed with diethyl ether (50 mL x 2) followed by Soxhlet extraction with acetonitrile for 12 h. Elemental analysis: For $(\text{C}_{26}\text{H}_{22}\text{BLiO}_4)_n$, calculated: C, 75.03; H, 5.33; B, 2.60; Li, 1.67%. Found: C, 59.05; H, 6.45; B, 2.48; Li, 2.09. NMR spectra in CD_3OD : ^{11}B , 3.54 ppm; ^1H , 8.54, 6.56–6.54–6.32 (multiple & broad), 1.80 (broad), 1.40 ppm (broad). FTIR [ATR(KBr) cm^{-1}]: $\nu = 3400, 2919, 1612, 1513, 1447, 1358, 1241, 1172, 1013, 828$.

2.3. Gel polymer preparation

55 mg of LiPVPPB and 165 mg of PVDF-HFP were added into 7 mL of DMF, which was stirred at 60°C to produce a homogeneous solution. Subsequently, the LiPVPPB/PVDF-HFP membrane was prepared via a doctor-blading process and dried in a vacuum oven at 80°C . Finally, the membrane was transferred into a glove box and immersed in an EC: PC (1:1) solution for soaking.

2.4. Methods

The infrared spectrum of LiPVPPB was recorded with the SHIMADZU 8400 FTIR spectrophotometer with a frequency range of $600\text{--}4000 \text{ cm}^{-1}$. The ^1H NMR spectra and the ^{11}B NMR spectra of the polymer were taken on AVANCE 500, BRUKER at room temperature. The sample was dissolved in CD_3OD . $\text{BF}_3\cdot\text{Et}_2\text{O}$ was used as a reference for the ^{11}B NMR. Thermal gravimetric analysis of the compound was investigated using SDT TA Instruments 2960 Simultaneous DTA-TGA at a heating rate of $10^\circ\text{C min}^{-1}$ under a constant nitrogen flow of 110 mL min^{-1} from room temperature to 800°C . The surface morphology of the polymer and its membrane were imaged using EOLJSM-6701F field emission scanning electron microscopy (FE-SEM). The samples were prepared by platinum-sputtering (20s, 20 mA).

The ionic conductivity of the prepared LiPVPPB/PVDF-HFP blend membrane was measured using electrochemical impedance spectroscopy (EIS) with a potentiostat-galvanostat AUTOLAB model PGSTAT12/30/302 over a frequency range of 1 Hz–4 MHz with an oscillating voltage of 5 mV. The polymer electrolyte was sandwiched between two stainless steel electrodes in a cylindrical device. The device was heated at 60°C for overnight prior to analysis to ensure a good contact between the electrolyte and the electrodes. The linear sweep voltammetry was carried out using the CHI workstation at a rate of 1 mV s^{-1} in the range of 2.5 to 6 V at room temperature. A lithium foil was employed as the counter and

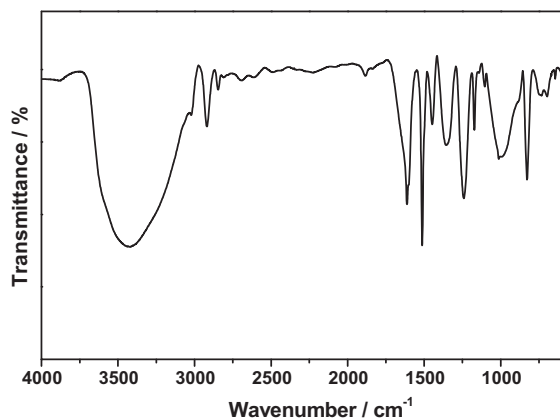


Fig. 1. The FT-IR spectra of lithium poly (4-vinylphenol) phenolate borate (LiPVPPB).

reference electrode. The lithium ion transference number was measured using the method proposed by Evans[44] by sandwiching the membrane with two lithium foil electrodes. After the initial resistance was measured by EIS, a potential of 10 mV was applied until a steady state was reached. Subsequently, the final resistance was measured by EIS again.

To evaluate the battery performance of the prepared electrolyte membrane, standard coin cells (CR2025) based on the LiPVPPB/PVDF-HFP composite membrane were assembled using a lithium foil anode and a carbon-coated LiFePO_4 cathode inside a glove box. The discharge capacity was measured on a multichannel battery testing instrument Arbin BT-2000. The formulation of the cathode was 75wt% LiFePO_4 , 10wt% PVDF, 10wt% acetylene black and 5wt% lithium sulphonamide.

3. Results and discussions

3.1. Synthesis and characterization of lithium poly (4-vinylphenol) phenolate borate (LiPVPPB)

LiPVPPB was obtained using the synthetic procedure shown in Scheme 1. The IR spectrum of the compound, shown in Fig. 1, exhibits the characteristic absorption bands (1358 cm^{-1} , 1241 cm^{-1} , 1172 cm^{-1} , 1013 cm^{-1}) associated with the B-O stretching modes.[45] The broad peak in the proximity of 3400 cm^{-1} is attributed to the residual hydroxyl group of the starting material, poly (4-vinylphenol). The peak at 2919 cm^{-1} corresponds to the stretching mode of the $-\text{CH}_2-$ group. The

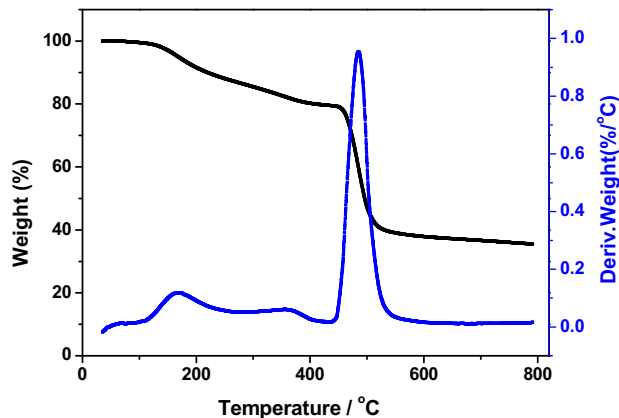


Fig. 3. The TGA thermograph of LiPVPPB (nitrogen, $10^\circ\text{C. min}^{-1}$, RT to 800°C).

frequencies at 1612 cm^{-1} , 1513 cm^{-1} and 1447 cm^{-1} reflect the characteristic vibration modes of benzene ring framework.

The ^1H NMR (left) and the ^{11}B NMR (right) spectra of LiPVPPB in CD_3OD are displayed in Fig. 2. The two groups of the broad peaks shown ^1H NMR correspond to the benzene rings (low field) and the ethylene groups (high field), respectively, verifying the proposed polymer structure in Scheme 1. The signal at 8.54 ppm is attributed to the residual hydroxyl group of poly (4-vinylphenol) entangled with the targeted polymer structure in the polymer matrix, consistent with the FT-IR spectrum shown in Fig. 1. The chemical shift of boron in the ^{11}B NMR spectrum further confirms the formation of the sp^3 configuration of boron atoms in the polymeric electrolyte. [46]

The TGA curve of LiPVPPB under nitrogen is shown in Fig. 3. Two stages of degradation are observed. One begins roughly at 120°C and extends to 400°C (approximately 20% of the weight loss) and the other starts at 450°C (approximately 40% of the weight loss). The first 20% weight loss is attributed to the loss of one phenolate group since boron atoms can also form a planar structure with a sp^2 configuration. The additional 40% weight loss is likely due to the loss of another two phenolate groups. The TGA measurement suggests that LiPVPPB is thermally stable in the temperature range of battery operation.

The FE-SEM images of the LiPVPPB polymer and the LiPVPPB/PVDF-HFP blend membrane are shown in Fig. 4. The LiPVPPB complex displays a flake-like morphology. The blend membrane prepared by the doctor-blading method is uniform and dense and thus enables a good contact with the electrodes.

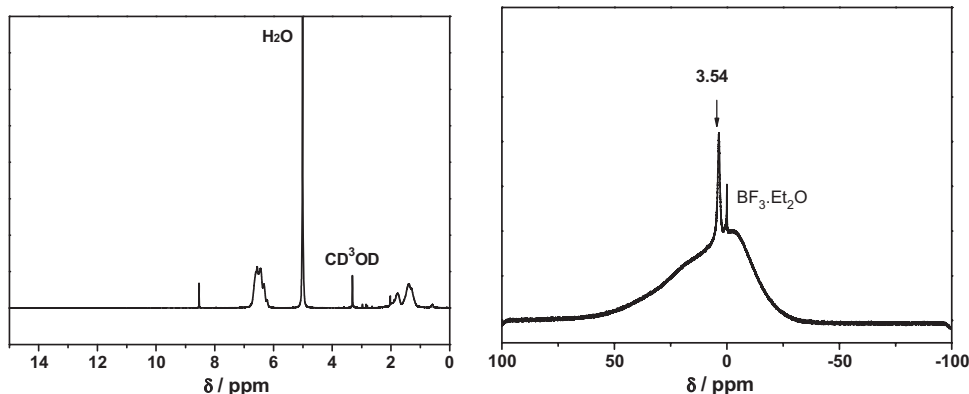


Fig. 2. The ^1H NMR (left) and ^{11}B NMR (right) spectra of LiPVPPB in CD_3OD .

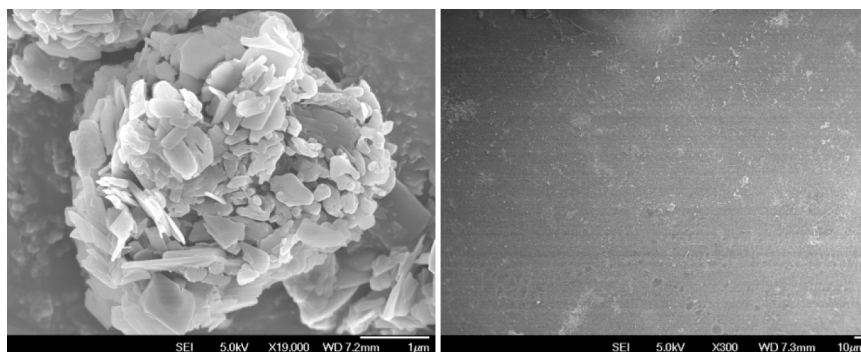


Fig. 4. The FE-SEM images of LiPVPPB (left) and LiPVPPB/PVDF-HFP membrane (right).

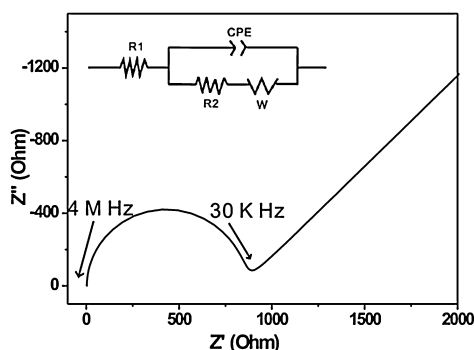


Fig. 5. The EIS plot of the LiPVPPB/PVDF-HFP composite membrane at room temperature.

3.2. Ionic conductivity

The ionic conductivity of the polymer electrolyte was calculated using the equation defined by:

$$\sigma = \frac{l}{Ra}$$

where σ is the ionic conductivity, l is the thickness of the electrolyte membrane, R is the bulk resistance and a is the surface area. The EIS plot of the electrolyte at room temperature is depicted in Fig. 5. The Nyquist plot exhibits a suppressed semicircle on the axis followed by a straight line. The semicircle portion (observed at high frequencies) corresponds to the electron-transfer-limited process, while the straight line (in the low frequency range) represents the diffusion-limited process.[47]

The ionic conductivity dependence on the inverse of temperature ($80^\circ\text{C} - 25^\circ\text{C}$, downwards) is depicted in Fig. 6, which displays a typical Arrhenius behavior. In contrast to what is observed in liquid electrolytes and lithium-salts-based polymer electrolytes, the conductivity is less sensitive to temperature, which is common among single-ion polymer electrolytes.[20–22] The relatively narrow variation of the conductivity in the temperature range is attributed to the mechanical coupling between the ion transport and the mobility of the polymer host according to the free volume law.[10,17] The measured ionic conductivities are $4.4 \times 10^{-4} \text{ S cm}^{-1}$ at 298 K and $1.5 \times 10^{-3} \text{ S cm}^{-1}$ at 313 K, respectively. The high ionic conductivity is chiefly due to the extensive electron delocalization of the anionic charge in the polymer framework, leading to significantly enhanced mobility of lithium ions.

3.3. Electrochemical stability

The electrochemical stability of the polymer electrolyte membrane was analyzed by linear sweep voltammetry using

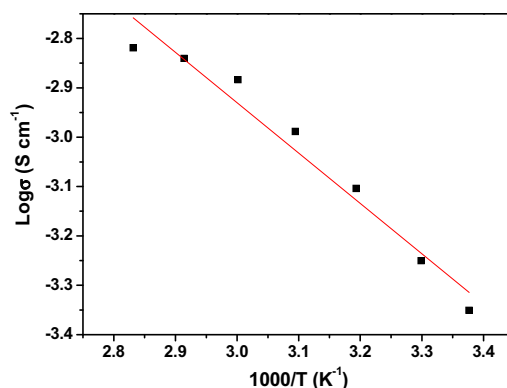


Fig. 6. The Arrhenius plot of log (ionic conductivity) versus inverse absolute temperature.

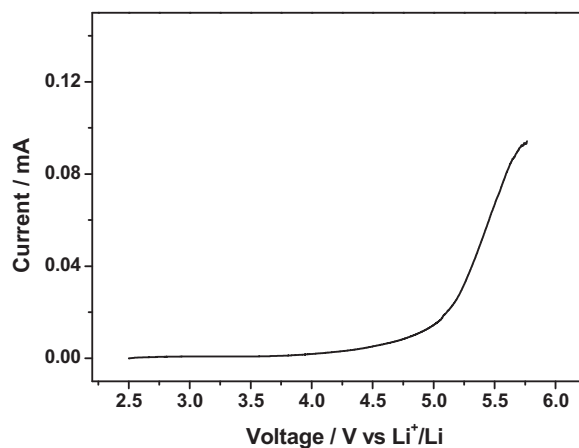


Fig. 7. The linear sweep voltammetry of the LiPVPPB/PVDF-HFP membrane (scan rate 1 mV s^{-1}).

Li/composite membrane/stainless steel cells. The measurement was performed between 2.5 V and 6.0 V (versus Li^+/Li) at a scan rate of 1 mV s^{-1} as shown in Fig. 7. The polymer electrolyte was found to be stable up to 4.5 V, indicating a broad electrochemical window. This result suggests that the LiPVPPB/PVDF-HFP blend membrane may be suitable for battery operation with most commonly used cathode materials[6], such as LiFePO_4 (3.8 V), Mn_xO_y (4.2 V).

3.4. Lithium-ion transference number

The lithium-ion transference number (t_{Li^+}) was measured by sandwiching the blend membrane between two nonblocking lithium metal electrodes.[18,48] The transference number was

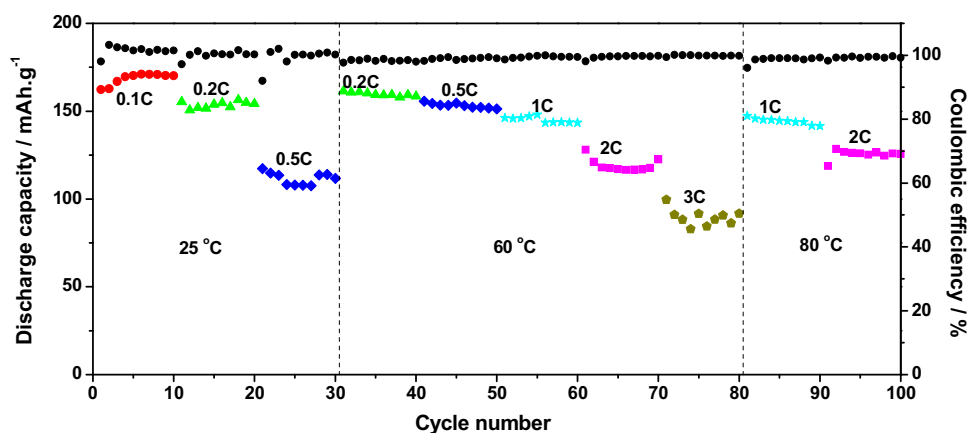


Fig. 8. The cycle performance of the Li/LiPVPPB/PVDF-HFP/LiFePO₄ cell at different temperatures and different C rates.

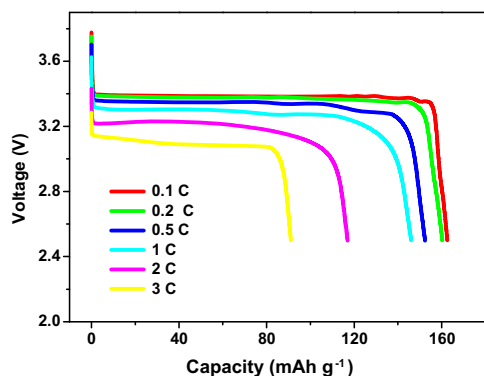


Fig. 9. Discharge profiles obtained at 60 °C at different C rates from 0.1 C to 3 C.

evaluated via combination of DC potentialstatic measurements and AC impedance spectroscopy based on the procedure proposed by Evans et al. [44] using the following equation:

$$t_{Li}^+ = \frac{I_s(\Delta V - I_o R_o)}{I_o(\Delta V - I_s R_s)}$$

where ΔV is the DC potential applied across the cell, R_o and R_s are the initial and the steady-state resistances of the passivation layer on the electrodes before and after DC polarization, and I_o and I_s are the initial and the steady-state currents. The t_{Li}^+ value was calculated to be 0.91 at room temperature, which is substantially higher than the values of dual-ion based electrolytes. The result indicates that the material is indeed a typical single-ion conductor.

3.5. Battery performance

To evaluate the electrochemical performance of the blend polymer electrolyte membrane, we assembled a coin cell battery device using a lithium foil as the anode and a carbon-coated LiFePO₄ as the cathode, which has a theoretical discharge capacity of 170 mAh g⁻¹. [49] Cycle tests were conducted at several temperatures with various C-rates. The results are shown in Fig. 8. Remarkably, the battery exhibits a high reversible discharge capacity over a wide temperature range from room temperature to 80 °C and even at higher power rates with the coulombic efficiency approaching 100% over the different power rates.

Fig. 9 shows the discharge profiles of the Li/LiPVPPB/PVDF-HFP/LiFePO₄ battery at 60 °C at different power rates. The discharge capacities of the battery were measured to be 165 mAh g⁻¹ at 0.1 C, 160 mAh g⁻¹ at 0.2 C, 150 mAh g⁻¹ at 0.5 C, 140 mAh g⁻¹ at 1 C, 120 mAh g⁻¹ at 2 C and 90 mAh g⁻¹ at 3 C, corresponding to 97%, 94%,

88%, 82% and 53% of the theoretical value, respectively. The reason for the high retention of the capacity at 1 C and 2 C rates is attributed to the high lithium transference number, which limits the formation of concentration gradient. [50]

4. Conclusions

A new single-ion polymer electrolyte (LiPVPPB) with boron atoms covalently bonded to a polystyrene backbone with a sp³ electronic configuration was successfully synthesized and the structures and properties of the polymer and its composite membrane with PVDF-HFP were thoroughly characterized via a range of spectroscopic techniques including thermogravimetric analysis, impedance spectroscopy and linear sweep voltammetry. The polymer electrolyte exhibits good thermal stability (up to 120 °C) and electrochemical stability (up to 4.5 V) with remarkably high ionic conductivity of 4.4×10^{-4} S cm⁻¹ at room temperature. The high ion conduction is attributed to the extensive electron delocalization through π -conjugation surrounding the sp³ boron atoms, resulting in the improved stability of anions and the enhanced mobility of lithium ions. The measured lithium ion transference number is close to unity (0.91), substantially higher than the value of typical dual-ion based electrolytes. The wide electrochemical window enables the electrolyte membrane to be potentially useful for batteries with metal oxides as cathode. These materials require electrolytes to be capable of sustaining high voltage upon charge and discharge. Furthermore, the battery performance of the SIPE membrane was evaluated using an assembled coin cell of Li/LiPVPPB/PVDF-HFP/LiFePO₄. Remarkable performance of the battery in a broad temperature range from room temperature to 80 °C with nearly 100% coulombic efficiency was demonstrated. Our results suggest that SIPE materials are highly promising for safe and high power applications of Li-ion batteries.

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